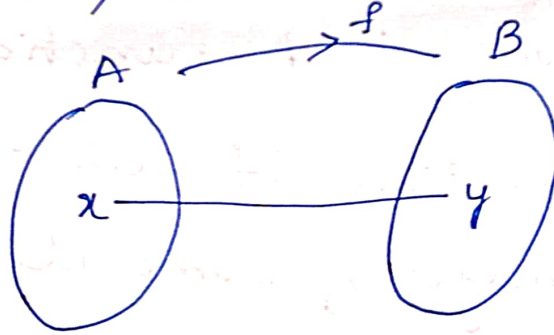


Functions

Let A and B be two non empty sets. A function or a mapping f from A to B is a rule which associates every element of A with a unique element of B and is denoted by $f: A \rightarrow B$.

If $f: A \rightarrow B$ is a function, then A is called domain, B is called co-domain of f .



If $x \in A$ is associated with a unique element $y \in B$ by the function f , then y is called image of x under f and is denoted by $y = f(x)$. Also x is called the

pre-image of y under f $[x = f^{-1}(y)]$

The set consisting of all images is called Range of f . It is denoted by $f(A)$

$$\text{So } f(A) = \{f(x) \in B \mid x \in A\} \subseteq B.$$

Types of functions

Into function : If the function

$f: A \rightarrow B$ is such that there is at least one element in B which is not the image of any element in A , then f is called into function

2. onto function : A function $f: A \rightarrow B$

is onto function if all the elements in B are images of some elements of A .

$$\text{So } \forall y \in B, \exists x \in A \ni y = f(x)$$

one one function : A function $f: A \rightarrow B$

is one one if different elements of A have different images in B

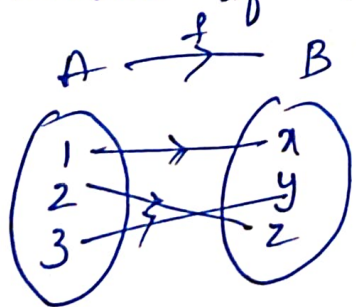
ie $\forall x_1, x_2 \in A, f(x_1) = f(x_2)$
 $\Rightarrow x_1 = x_2$

(4) Many one function: A function $f: A \rightarrow B$ is many one function if two or more elements in A have same image in B .

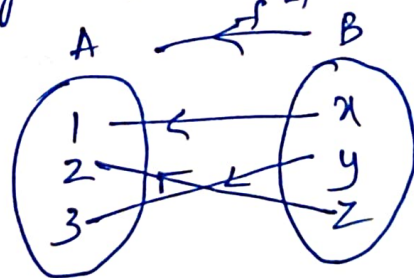
(5) Bijective function: A function $f: A \rightarrow B$ is bijective if f is both one-one and onto.

(6) Inverse function: If $f: A \rightarrow B$ is a bijective function then the mapping $f^{-1}: B \rightarrow A$ which associates each element $y \in B$ to a unique element $x \in A$ such that $f(x) = y$ or $x = f^{-1}(y)$ is called

inverse of the mapping $f: A \rightarrow B$



$f(1) = x, f(2) = z, f(3) = y$



$f^{-1}(x) = 1, f^{-1}(y) = 3, f^{-1}(z) = 2$

Composite functions

Let $f: A \rightarrow B$ and $g: B \rightarrow C$ be two functions. Then composition of functions f and g is denoted by $g \circ f$ is the function $g \circ f: A \rightarrow C$ and is defined by $(g \circ f)(x) = g[f(x)]$

Similarly $(f \circ g)(x) = f[g(x)]$

Identity function

A function $f: A \rightarrow A$ defined by $f(x) = x$ for every $x \in A$ is called identity function on A . It is also denoted by I_A .